

12 Chemistry Memorandum

Prelim 2024

- 1.1 D ✓ 1.2 B ✓ 1.3 C ✓ 1.4 D ✓ 1.5 B ✓
 1.6 D ✓ 1.7 B ✓ 1.8 A ✓ 1.9 B ✓ 1.10 D ✓
 C
 10 × 2 = 20

Question 2

- 2.1 The force in a solid or liquid between the molecules of a compound ✓✓ 2
- 2.2 1 B ✓
 2. C ✓
 3. ethyl ethanoate ✓ or propyl methanoate 3
- 2.3.1 esterification ✓
 2. H_2SO_4 / sulphuric acid ✓
 3. flammability ✓ 3
- 2.4 $M(C_3H_6O) = 3(12) + 6 + 16 = 58$
 $n = \frac{116}{58} = 2$
 True formula: $C_6H_{12}O_2$
 C of acid: $\therefore = C_2H_4O_2$
 $\Rightarrow$ ethanoic acid ✓
- 2.5.1 pent-2-yne ✓

$$\begin{array}{c} H & H & C & H \\ | & | & \vdots & | \\ H-C & -C- & C-H \\ | & | & | \\ H & O & H \\ & | & \\ & H & \end{array}$$

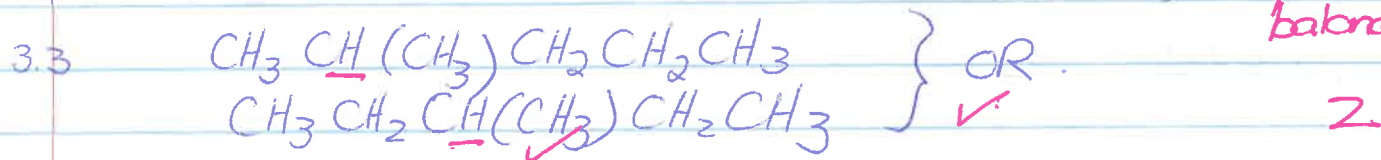
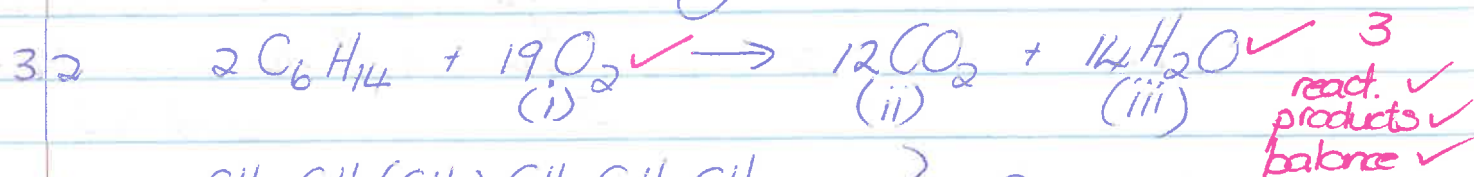
 2.5.2 pentan-3-one ✓
 methyl propan-2-ol. ✓
 2

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(2)

Question 3.

3.1 Elimination, cracking ✓ 2

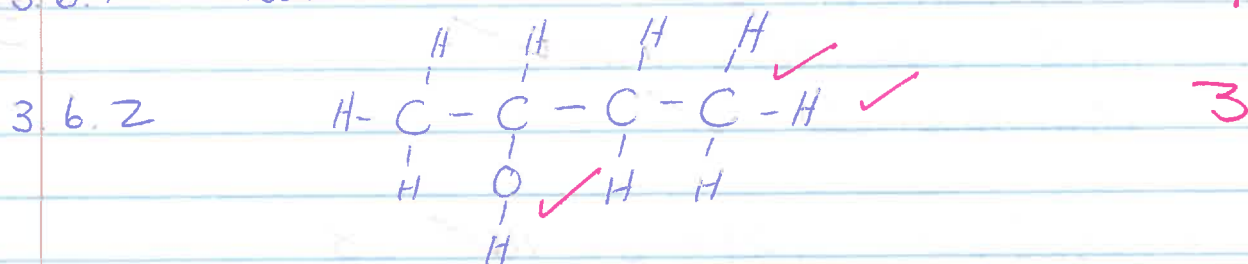


3.4 but-2-ene ✓ or but-1-ene 1

3.5.1 hydrohalogenation / hydrochlorination 1

3.5.2 No water / anhydrous HX ✓ 1

3.6.1 substitution ✓ 1



3.6.3 IV NaOH ✓
V salt: NaCl ✓ 2

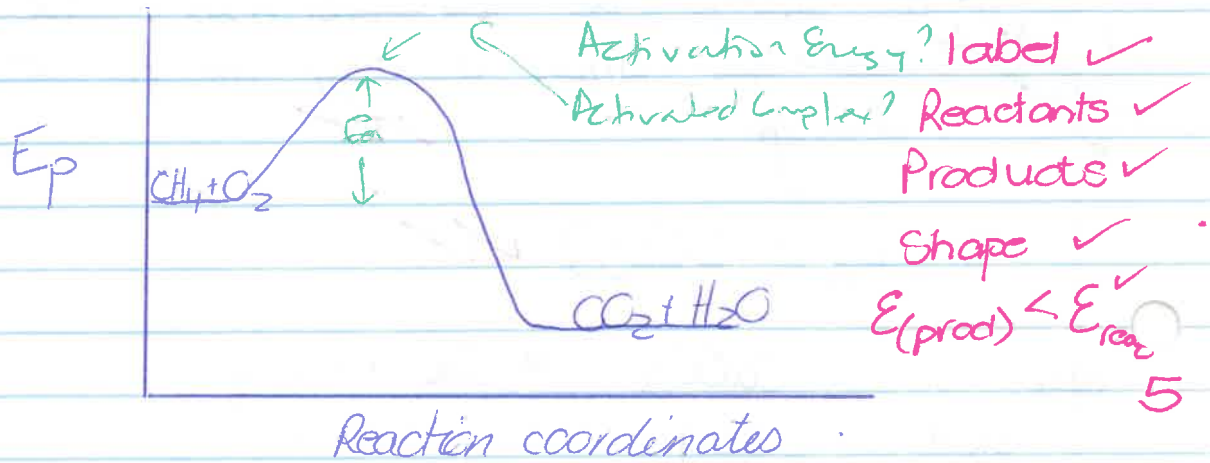
3.6.4 Diluted NaOH ✓ moderate heat ✓ 2

[18].

Question 4.

4.1 a chemical reaction that releases energy/
the energy of the products is lower than the
reactants ✓✓ 2

4.2



4.3 Increased O_2 molecules, collisions per unit reaction rate ✓, more effective ✓, higher ✓, in a certain time ✓.
 4.4.1 ?

4.4.2 Adding a catalyst ✓ (4)

4.4.3 catalyst influence forward + reverse rxn equally ✓✓ 2

4.4.4 No change ✓ REMAIN THE SAME. 1

4.4.1 The rate of the reverse reaction ✓ 1

[16]

Question 5

5.1.1. The number of atoms in the reaction stays the same, no matter escapes / is constant. 2
 2 When a change (P, V, T) is made on a system in equilibrium, the system will react in such a way as to counteract the change. 2
Give to state from guidelines 2

3. $K_c = [CO_2]$ ✓✓ 2



Start 35g
 Change 12g ✓
 Equilibrium 23g

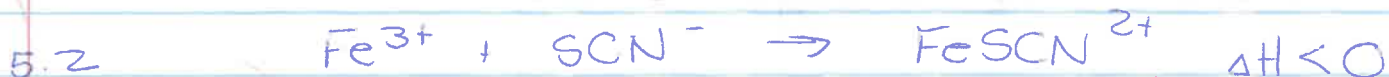
12g $CaCO_3$ $n = \frac{m}{M} = \frac{12}{100} = 0,12 \text{ mol}$ ✓



0,12 mol ✓

$C = \frac{n}{V} = \frac{0,12}{0,5} = 0,24 \text{ mol/dm}^3$ ✓

$K_c = 0,2$ ✓ 5



The forward reaction is exothermic, as temp. increases when the ions are added. An increase in temperature will favour the endothermic reaction, the reverse reaction is endothermic, will favour $Fe^{3+} + SCN^-$ ions, so the colour will turn yellow. ✓

4
 [15]

Question 6.

6.1 It ionises completely in water to form a high concentration of H_3O^+ ions ✓✓ 2

6.2 $0,2 \text{ mol dm}^{-3} \text{ NaOH}$

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] \quad \checkmark$$

$$1 \times 10^{-14} \checkmark = [\text{H}_3\text{O}^+][0,2] \quad \checkmark$$

$$[\text{H}_3\text{O}^+] = 5 \times 10^{-14} \text{ mol dm}^{-3} \quad \checkmark$$

4.

6.3.1 neutralisation point ✓

$\text{pH} = 7$ ✓

2

2. bromothymol blue ✓

Strong acid + strong base ✓

2

6.4 $\text{HCl} + \text{NaHCO}_3$

V	$0,5 \text{ dm}^3$	$0,8 \text{ dm}^3$
C	$0,1 \text{ mol dm}^{-3}$	$0,25 \text{ mol dm}^{-3}$

Start: $n = CV$ ✓

$$= 0,1 \times 0,5$$

$$= 0,05 \text{ mol} \quad \checkmark$$

$$0,8 \times 0,25$$

$$n = 0,2 \text{ mol} \quad \checkmark$$

Ratio

Limits $0,05$

$$\text{Left } 0,2 - 0,05$$

$$= 0,15 \text{ mol}$$

$$C = \frac{n}{V} = \frac{0,15}{0,13} = 1,15 \text{ mol dm}^{-3}$$

$$C = \frac{n}{V} = \frac{0,015}{1,3} = 0,011$$

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$$

$$1 \times 10^{-14} = [\text{H}_3\text{O}^+][0,011]$$

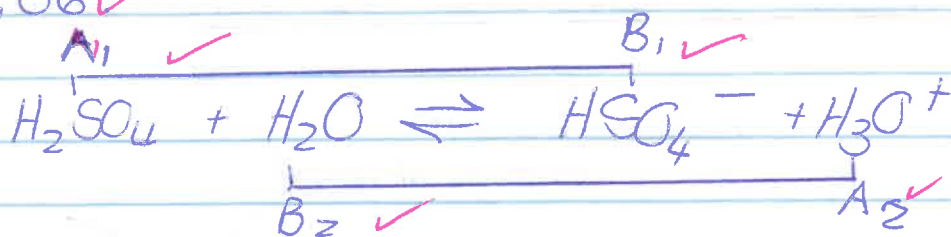
$$[\text{H}_3\text{O}^+] = 8,66 \times 10^{-13} \quad \checkmark$$

8

$$\begin{aligned}
 \text{pH} &= -\log [\text{H}_3\text{O}^+] \quad \checkmark \\
 &= -\log [1,8,66 \times 10^{-13}] \quad \checkmark \leftarrow \text{Charge} \\
 &= 12,06 \quad \checkmark
 \end{aligned}$$

4

6.5



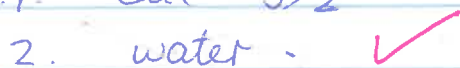
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Question 7



2



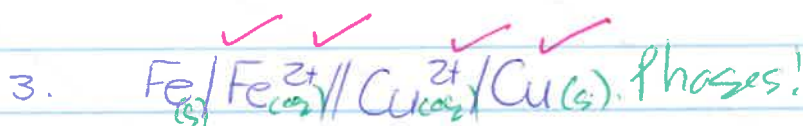
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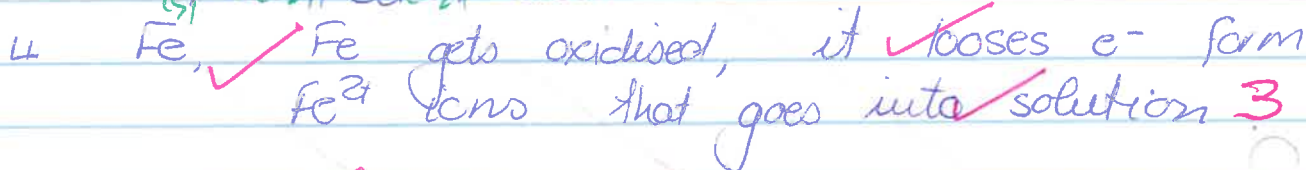
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2



4



3

5. Yes: $\checkmark$ It does not form a precipitate.
 Cl^- ions moves to Fe^{2+} half, as $[\text{Fe}^{2+}]$ increases to neutralise excess charge
 K^+ ions moves towards Cu^{2+} half cell to replace the reduced Cu^{2+} ions and to keep neutral charges $\checkmark$

4

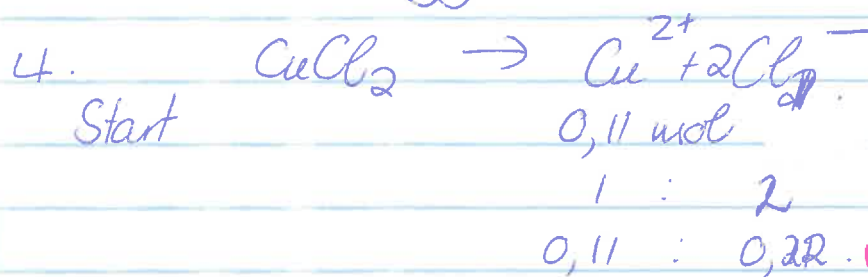
$$\begin{aligned}
 6. \text{EMF} &= E^\theta(\text{cath}) - E^\theta(\text{anode}) \quad \checkmark \\
 &= +0,34 - (-0,44) \quad \checkmark \\
 &= 0,78 \text{ V} \quad \checkmark
 \end{aligned}$$

3

[20]

Question 8

- 8.1.1 The solid has no free ions to move, as it is packed in a solid lattice structure 3
2. Anode 2
3. The $[\text{CuCl}_2]$ decreases, if it gets too low water will be oxidised to O_2 . It is a stronger reducing agent than Cl^- and less energy will be needed. 3



no of Cl^- ions
 $n = \frac{\text{no}}{N_A}$

0,22 = $\frac{\text{no}}{6,02 \times 10^{23}}$

no = $1,3244 \times 10^{23} \text{ Cl}^- \text{ ions}$

8.1.5 no of e^- : $1,3244 \times 10^{23} e^- \text{ transferred}$
 $Q = 1,3244 \times 10^{23} \times 1,6 \times 10^{-19}$
 $21190,4 \text{ C}$
 $Q = It \quad \frac{21190,4}{600} = 35,31 \text{ A}$

8.1.6 non spontaneous ✓

8.2.1 Th